

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer 2024

**Course: B. Tech. Branch : Electronics & Telecommunication Engineering /
Electronics & Communication Engineering/ Electronics Engineering Semester : IV
Subject Code: BTETPE405A / BTEXPE405A**

Subject Name: Numerical Methods and Computer Programming

Max Marks: 60

Date: 24/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

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(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

A) Explain the term with help of suitable example

CO-1

6

1. Round-off error

2. Relative error

3. Truncation error

B) Explain the term

CO-1

6

1. Error Estimation

2. Error propagation

C) An approximate value of π is given by $x_1 = 22/7 = 3.142857$ and its true value is $x = 3.1415926$ find the relative error

CO-2

6

Q.2 Solve Any Two of the following.

12

A) Find the real root of the equation $e^x \cos x = 1.2$ up to 3 decimal accuracy assume root lies between 0 and 1 . Use Bisection Method

CO-2

6

B) Use Newton Raphson method find the root $x^4 - x - 10 = 0$ which is near to $x = 2$

CO-2

6

C) Solve the following system by Gauss Seidel Method

CO-2

6

$$10x - 5y - 2z = 3$$

$$4x - 10y + 3z = -3$$

$$x + 6y + 10z = -3$$

Q. 3 Solve Any Two of the following.

12

A) Using stirling's formula, find $y(35)$ from the following table

CO-2

6

x	20	30	40	50
y	512	439	346	243

B) Using Newton Divided difference formula find the values of $f(2)$, $f(8)$, $f(15)$ from the following table

CO-2

6

x	4	5	7	10	11	13
f(x)	48	100	294	900	1210	2028

C) Find $y(10)$ from the following table , by using Lagrange's interpolation formula

CO-2

6

x	5	6	9	11
y	12	13	14	16

Q.4 Solve Any Two of the following.

12

A) Evaluate $\int_0^3 2x - x^2 dx$, taking 6 intervals by Trapezoidal rule

CO-2

6

B) Evaluate $\int_{1.3}^{2.5} x + 2\sin x dx$, taking 12 strips by Simpson's 3/8th rule

CO-2

6

C) Find y (2.2) by using Euler's method for $\frac{dy}{dx} = xy^2$ where y (2) = 1.
consider h = 0.1

CO-4

6

Q.5 Solve Any Two of the following.

12

A) Explain difference between constructor and destructors with example

CO-2

6

B) Explain function overloading

CO-2

6

C) Explain application of OOP

CO-2

6

***** End *****